Data-Backed Journal-Style Replication of *Ambition Is Nothing Without Focus*

Codex journal-replication workflow

2026-04-04

Abstract

This manuscript expands the local replication folder for Ghosh, Martin, Pennings, and Wezel (2014) into a long journal-style paper using only processed datasets already present in the workspace. The current local batch contains 489,193 observations across 200 entities, and the batch headline estimate is -0.0022 for breadth x org_recomb_exp_log with p = 0.0076. The local rerun strongly preserves the negative breadth-by-experience interaction and the negative three-way interaction with organizational focus. The draft imitates the structure of the source article while documenting the exact places where local replication decisions depart from direct source facts.

Keywords: recombination breadth; organizational focus; inventive impact; patent citations; local reproduction

1 Paper-First Reading Protocol

The workflow for this manuscript is intentionally ordered. The source paper is read first, the required empirical objects are identified second, the local files are verified third, and only then is the replication paper written from the checked evidence chain.

Step	Action	Evidence
1	Read the original paper before touching the local estimates.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/Project 1_ Ghosh et al. 2014 replication/Ghosh-Martin-Pennings-Wezel-OS-25-(2)-2014.pdf
2	Extract the source paper's theoretical logic, sample definition, variable definitions, and estimand.	Ambition Is Nothing Without Focus: Compensating for Negative Transfer of Experience in R&D
3	Match each required empirical object to a local executable file and verify that the file exists.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/Data_Analysis/code/Regression/output/gvkey_ — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/ghosh-martin-pennings-wezel-2014-ambition-focus/results/ghosh_2014_model_coefficients.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/ghosh-martin-pennings-wezel-2014-ambition-focus/results/ghosh_2014_sample_summary.csv

Step	Action	Evidence
4	Write the replication paper only from the checked local files and calculated outputs.	The manuscript, markdown notes, and PDF are generated from the verified file set.

Ordered protocol used for this replication paper.

2 Required Data and Local File Manifest

This section answers the basic replication question of where the data are. Each empirical object is mapped to a concrete local file before the estimates are interpreted. Workspace root: C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions

Stage	NeededFor	FilePath	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	Project 1_Ghosh et al. 2014 replication/Ghosh-Martin-Pennings-Wezel-OS-25-(2)-2014.pdf	Yes	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	Data_Analysis/code/Regression/outputs/cleaned_dataset	Yes	Executable local reproduction

Stage	NeededFor	FilePath	Exists	Status
Coefficient export	Recover the reported local coefficients and p-values	patent literature/daily_replications/2026-04-04/ghosh-martin-pennings-wezel-2014-ambition-focus/results/ghosh_2014_model_coefficients.csv	Yes	Executable local result file
Sample summary	Audit sample construction and raw grouped moments	patent literature/daily_replications/2026-04-04/ghosh-martin-pennings-wezel-2014-ambition-focus/results/ghosh_2014_sample_summary.csv	Yes	Executable local result file

Exact local files used by the manuscript.

3 Introduction

The source paper asks whether broad recombination becomes less productive when it leans on familiar routines, unless organizational focus offsets that negative transfer. The point of the present draft is not to pretend that every original source file has been recovered. It is to turn the executable local evidence into a paper that preserves theory, data architecture, variable mapping, and empirical claims in one place. The current local batch contains 489,193 observations across 200 entities, and the batch headline estimate is -0.0022 for breadth x org_recomb_exp_log with p = 0.0076. The local rerun strongly preserves the negative breadth-by-experience interaction and the negative three-way interaction with organizational focus.

4 Original Paper, Theory, and Published Estimand

The focal source article is [1]. The source paper is Ambition Is Nothing Without Focus: Compensating for Negative Transfer of Experience in R&D by Anindya Ghosh, Xavier Martin, Johannes M. Pennings, and Filippo Carlo Wezel in Organization Science . The source paper studies the interaction of breadth, organizational experience, and focus. Forward citations proxy inventive impact. The unit of analysis is the patent. A useful replication discipline is to keep source facts and replication decisions separate. Source facts describe what the paper says. Replication decisions describe what the local executable workflow must do because it is limited to processed datasets already present in the workspace.

- The source paper studies the interaction of breadth, organizational experience, and focus.
- Forward citations proxy inventive impact.
- The unit of analysis is the patent.

DirectSourceFact	LocalReplicationDecision
The interaction structure is the theoretical core.	The local rerun keeps the same interaction logic on the clean GVKEY-based panel.
Forward citations measure impact.	The local impact measure is the executable citation outcome already stored in the panel.
The paper works at the patent level.	The sample is narrowed to active S&P 500 utility patents with unambiguous assignment.

Direct source facts are kept separate from local replication decisions.

5 Local Data Inventory and Sample Reconstruction

The local rerun uses the clean GVKEY-anchored utility-patent panel already built in the workspace and keeps the patent-level interaction structure of the original paper. Every number in this manuscript is backed by local files in patent literature/daily_replications/2026-04-04/ghosh-martin-pennings-wezel-2014-ambition-focus or by the explicitly referenced processed regression panel.

- Clean GVKEY-anchored S&P 500 utility-patent panel from the local replication pipeline.
- PatentView-derived application year and citation outcomes.
- CRSP-Compustat linked firm-year filters and sector information already present in the workspace.

File	Role
gvkey_clean_reproduction_dataset.csv	Clean patent-level regression panel
ghosh_2014_model_coefficients.csv	Baseline coefficient dump
ghosh_2014_focus_table.csv	Headline interaction estimates across sample variants
ghosh_2014_sample_summary.csv	Sample-construction audit log

Local executable files used by this paper.

Step	Rows	UniquePatents	UniqueGVKEY	UniqueYears	AmbiguousPatents	Notes
regression_dataset_raw	499486.0000	495460.0000	204.0000	31.0000	NA	Patent-level regression dataset produced by Regression.R
after_active_and_patent_type_dedupe	499486.0000	495460.0000	204.0000	31.0000	3678.0000	Active panel duplicates on gvkey-year before tie-break: 0
sp500_utility_drop_ambiguous	489193.0000	489193.0000	200.0000	31.0000	0.0000	Final clean sample used for reproduction

Sample reconstruction table from the local results folder.

6 Variable Mapping, Derivations, and Replication Decisions

The variable map is the bridge between the source article and the executable local workflow.

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Inventive impact	Forward citations	inventive_impact	Use the local cleaned citation outcome already stored in the regression dataset.
Breadth	Centered breadth	breadth_c	Carry forward the centered breadth variable from the local regression output.
Organizational experience	Centered log prior organizational experience	org_recomb_exp_log_c	Carry forward the centered log experience variable from the same panel.
Organizational focus	Centered firm-year focus	org_focus_c	Use the contemporaneous focus measure already reconstructed in the local pipeline.
Controls	Prior-art citations, degree, field experience, scale, age	n_prior_art_cites; recomb_degree; field_recomb_exp_log; total_patents_same_year; firm_age	Reuse the local patent-level controls and fixed effects.

Variable mapping from source constructs to local measures. A patent-level count model with interactions among breadth, organizational recombination experience, and organizational focus, plus firm and year controls.

$$E[\text{Impact}_{pft} \mid Z_{pft}] = \exp\big(\beta_1 \text{Breadth}_{pft} + \beta_2 \text{OrgExp}_{pft} + \beta_3 \text{Focus}_{ft} + \beta_4 \text{Breadth}_{pft} \times \text{OrgExp}_{pft} + \beta_5 \text{OrgExp}_{pft} \tag{1}$$

7 Empirical Design and Identification Limits

The local design remains strongest when it preserves the original unit of analysis and interaction logic, even if some measurement layers are reconstructed with defensible analogues rather than identical source files.

- The local clean sample is paper-comparable rather than guaranteed identical to the archival source frame.
- Patent-firm matching and sample filters follow the executable local pipeline.
- Interpretation should emphasize mechanism recovery more than exact coefficient duplication.

8 Main Results

Breadth is estimated at 0.0030 with p = 0.8477 in model poisson_baseline. Organizational experience is estimated at 0.0706 with p = 0.0461 in model poisson_baseline. Organizational focus is estimated at -0.3609 with p = 0.3373 in model poisson_baseline. Breadth x org. experience is estimated at -0.0022 with p = 0.0076 in model poisson_baseline. Org. experience x focus is estimated at 0.3219 with p = 0.0703 in model poisson_baseline. Breadth x org. experience x focus is estimated at -0.1705 with p = 0.0152 in model poisson_baseline. Breadth is estimated at 0.0147 with p = 0.3159 in model logit_top5. Organizational experience is estimated at 0.0988 with p = 0.0256 in model logit_top5. Organizational focus is estimated at -0.8737 with p = 0.5200 in model logit_top5. Breadth x org. experience is estimated at -0.0118 with p = ¡0.001 in model logit_top5.

Org. experience x focus is estimated at 0.0361 with p = 0.9680 in model logit_top5. Breadth x org. experience x focus is estimated at -0.2765 with p = 0.3557 in model logit_top5. Breadth is estimated at -0.0099 with p = 0.6686 in model logit_top1. Organizational experience is estimated at 0.1380 with p = 0.1395 in model logit_top1. Organizational focus is estimated at -0.9087 with p = 0.8056 in model logit_top1. Breadth x org. experience is estimated at -0.0132 with p = 0.0184 in model logit_top1. Org. experience x focus is estimated at -0.5147 with p = 0.8763 in model logit_top1. Breadth x org. experience x focus is estimated at -0.6392 with p = 0.5361 in model logit_top1.

model	Term	estimate	std_error	p_value
poisson_baseline	Breadth	0.0030	0.0158	0.8477
poisson_baseline	Organizational experience	0.0706	0.0354	0.0461
poisson_baseline	Organizational focus	-0.3609	0.3761	0.3373
poisson_baseline	Breadth x org. experience	-0.0022	0.0008	0.0076
poisson_baseline	Org. experience x focus	0.3219	0.1779	0.0703
poisson_baseline	Breadth x org. experience x focus	-0.1705	0.0702	0.0152
logit_top5	Breadth	0.0147	0.0147	0.3159
logit_top5	Organizational experience	0.0988	0.0442	0.0256
logit_top5	Organizational focus	-0.8737	1.3579	0.5200
logit_top5	Breadth x org. experience	-0.0118	0.0023	0.0000
logit_top5	Org. experience x focus	0.0361	0.8998	0.9680
logit_top5	Breadth x org. experience x focus	-0.2765	0.2994	0.3557

model	Term	estimate	std_error	p_value
logit_top1	Breadth	-0.0099	0.0231	0.6686
logit_top1	Organizational experience	0.1380	0.0934	0.1395
logit_top1	Organizational focus	-0.9087	3.6923	0.8056
logit_top1	Breadth x org. experience	-0.0132	0.0056	0.0184
logit_top1	Org. experience x focus	-0.5147	3.3074	0.8763
logit_top1	Breadth x org. experience x focus	-0.6392	1.0331	0.5361

Key coefficient table used in the main text.

Term	Model	Estimate	PValue	Signal	Reading
Breadth	poisson_baseline	0.0030	0.8477	Weak	Positive estimate with p = 0.8477.
Organizational experience	poisson_baseline	0.0706	0.0461	Moderate	Positive estimate with p = 0.0461.
Organizational focus	poisson_baseline	-0.3609	0.3373	Weak	Negative estimate with p = 0.3373.

Term	Model	Estimate	PValue	Signal	Reading
Breadth x org. experience	poisson_baseline	-0.0022	0.0076	Strong	Negative estimate with p = 0.0076.
Org. experience x focus	poisson_baseline	0.3219	0.0703	Suggestive	Positive estimate with p = 0.0703.
Breadth x org. experience x focus	poisson_baseline	-0.1705	0.0152	Moderate	Negative estimate with p = 0.0152.
Breadth	logit_top5	0.0147	0.3159	Weak	Positive estimate with p = 0.3159.
Organizational experience	logit_top5	0.0988	0.0256	Moderate	Positive estimate with p = 0.0256.
Organizational focus	logit_top5	-0.8737	0.5200	Weak	Negative estimate with p = 0.5200.
Breadth x org. experience	logit_top5	-0.0118	0.0000	Strong	Negative estimate with p = ¡0.001.

Term	Model	Estimate	PValue	Signal	Reading
Org. experience x focus	logit_top5	0.0361	0.9680	Weak	Positive estimate with p = 0.9680.
Breadth x org. experience x focus	logit_top5	-0.2765	0.3557	Weak	Negative estimate with p = 0.3557.
Breadth	logit_top1	-0.0099	0.6686	Weak	Negative estimate with p = 0.6686.
Organizational experience	logit_top1	0.1380	0.1395	Weak	Positive estimate with p = 0.1395.
Organizational focus	logit_top1	-0.9087	0.8056	Weak	Negative estimate with p = 0.8056.
Breadth x org. experience	logit_top1	-0.0132	0.0184	Moderate	Negative estimate with p = 0.0184.
Org. experience x focus	logit_top1	-0.5147	0.8763	Weak	Negative estimate with p = 0.8763.

Term	Model	Estimate	PValue	Signal	Reading
Breadth x org. experience x focus	logit_top1	-0.6392	0.5361	Weak	Negative estimate with p = 0.5361.

Interpretive summary of the main coefficients.

Variable	N	Mean	SD	P25	Median	P75	Min	M
Inventive impact	489193.0000	6.2253	22.9068	1.0000	3.0000	6.0000	0.0000	1190.00
Breadth	489193.0000	-0.0064	5.2756	-3.0495	-1.0495	0.9505	-3.0495	129.95
Organizational experience	489193.0000	0.0070	1.3465	-0.7782	-0.7782	0.3204	-0.7782	6.86
Organizational focus	489193.0000	-0.0004	0.0301	-0.0088	-0.0066	-0.0017	-0.0104	0.98
Prior-art citations	489193.0000	18.9182	57.5082	5.0000	9.0000	16.0000	2.0000	2378.00
Recombination degree	489193.0000	0.5853	0.1778	0.4600	0.6122	0.7222	0.0282	0.96
Field experience	489193.0000	1.3943	1.8625	0.0000	0.6931	2.4849	0.0000	8.12
Same-year patents	489193.0000	1106.5533	1468.4128	228.0000	495.0000	1281.0000	1.0000	7321.00
Firm age	489193.0000	16.3219	7.3797	11.0000	17.0000	22.0000	1.0000	31.00

Descriptive statistics for the main variables.

Quantile	N	MeanRegressor	MeanOutcome
1.0000	229109.0000	-2.5926	4.9416
2.0000	76899.0000	-1.0495	5.3861

Quantile	N	MeanRegressor	MeanOutcome
3.0000	88598.0000	0.3530	5.8387
4.0000	94587.0000	6.7691	10.3793

Quantile summary linking the main regressor to the local outcome.

9 Comparison to the Published Paper and Remaining Gaps

This is the strongest executable match in the current batch. The negative breadth-by-experience interaction survives clearly, and the three-way interaction with focus remains negative and statistically meaningful, which means the central mechanism of the source paper is recovered in the local panel. The honest replication standard is to describe the gap without smoothing it over. A useful local paper tells the next researcher exactly which parts of the original design are recoverable now and which parts remain approximated.

10 Discussion and Conclusion

The manuscript turns a replication folder into a paper that can be uploaded, cited internally, and rerun later without guessing how the evidence was assembled. For Ghosh, Martin, Pennings, and Wezel (2014) the bottom line is: The local rerun strongly preserves the negative breadth-by-experience interaction and the negative three-way interaction with organizational focus.

A Appendix A: Source Inventory

Stage	NeededFor	FilePath	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	Project 1_ Ghosh et al. 2014 replication/Ghosh-Martin-Pennings-Wezel-OS-25-(2)-2014.pdf	Yes	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	Data_Analysis/code/Regression/output/ghosh_clean.rd	Yes	Executable local reproduction dataset
Coefficient export	Recover the reported local coefficients and p-values	patent literature/daily_replications/2026-04-04/ghosh-martin-pennings-wezel-2014-ambition-focus/results/ghosh_2014_model_coefficients.csv	Yes	Executable local result file
Sample summary	Audit sample construction and raw grouped moments	patent literature/daily_replications/2026-04-04/ghosh-martin-pennings-wezel-2014-ambition-focus/results/ghosh_2014_sample_summary.csv	Yes	Executable local result file

Source and local file manifest for the manuscript.

B Appendix B: Reading Protocol

Step	Action	Evidence
1	Read the original paper before touching the local estimates.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/Project 1_ Ghosh et al. 2014 replication/Ghosh-Martin-Pennings-Wezel-OS-25-(2)-2014.pdf
2	Extract the source paper's theoretical logic, sample definition, variable definitions, and estimand.	Ambition Is Nothing Without Focus: Compensating for Negative Transfer of Experience in R&D
3	Match each required empirical object to a local executable file and verify that the file exists.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/Data_Analysis/code/Regression/output/gvkey_ — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/ghosh-martin-pennings-wezel-2014-ambition-focus/results/ghosh_2014_model_coefficients.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/ghosh-martin-pennings-wezel-2014-ambition-focus/results/ghosh_2014_sample_summary.csv

Step	Action	Evidence
4	Write the replication paper only from the checked local files and calculated outputs.	The manuscript, markdown notes, and PDF are generated from the verified file set.

Paper-first protocol followed before writing the manuscript.

C Appendix C: Sample Reconstruction

Step	Rows	UniquePatents	UniqueGVKEY	UniqueYears	AmbiguousPatents	Notes
regression_dataset_raw	499486.0000	495460.0000	204.0000	31.0000	NA	Patent-level regression dataset produced by Regression.R
after_active_and_patent_type_coverage	499486.0000	495460.0000	204.0000	31.0000	3678.0000	Active panel duplicates on gvkey-year before tie-break: 0
sp500_utility_drop_ambiguous	489193.0000	489193.0000	200.0000	31.0000	0.0000	Final clean sample used for reproduction

Detailed sample reconstruction table.

D Appendix D: Variable Mapping

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Inventive impact	Forward citations	inventive_impact	Use the local cleaned citation outcome already stored in the regression dataset.
Breadth	Centered breadth	breadth_c	Carry forward the centered breadth variable from the local regression output.
Organizational experience	Centered log prior organizational experience	org_recomb_exp_log_c	Carry forward the centered log experience variable from the same panel.
Organizational focus	Centered firm-year focus	org_focus_c	Use the contemporaneous focus measure already reconstructed in the local pipeline.

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Controls	Prior-art citations, degree, field experience, scale, age	n_prior_art_cites; recomb_degree; field_recomb_exp_log; total_patents_same_year; firm_age	Reuse the local patent-level controls and fixed effects.

Detailed construct mapping.

E Appendix E: Descriptive Statistics

Variable	N	Mean	SD	P25	Median	P75	Min	M
Inventive impact	489193.0000	6.2253	22.9068	1.0000	3.0000	6.0000	0.0000	1190.00
Breadth	489193.0000	-0.0064	5.2756	-3.0495	-1.0495	0.9505	-3.0495	129.95
Organizational experience	489193.0000	0.0070	1.3465	-0.7782	-0.7782	0.3204	-0.7782	6.86
Organizational focus	489193.0000	-0.0004	0.0301	-0.0088	-0.0066	-0.0017	-0.0104	0.98
Prior-art citations	489193.0000	18.9182	57.5082	5.0000	9.0000	16.0000	2.0000	2378.00
Recombination degree	489193.0000	0.5853	0.1778	0.4600	0.6122	0.7222	0.0282	0.96
Field experience	489193.0000	1.3943	1.8625	0.0000	0.6931	2.4849	0.0000	8.12
Same-year patents	489193.0000	1106.5533	1468.4128	228.0000	495.0000	1281.0000	1.0000	7321.00
Firm age	489193.0000	16.3219	7.3797	11.0000	17.0000	22.0000	1.0000	31.00

Descriptive statistics appendix.

F Appendix F: Correlation Matrix

Variable	Inventive impact	Breadth	Organizational experience	Organizational focus	Prior-art citations	Recombination degree	Field experience	Same-year patents	Firm age
Inventive impact	1.000	0.297	-0.016	-0.011	0.502	0.014	-0.027	-0.018	0.005
Breadth	0.297	1.000	-0.209	-0.038	0.817	0.470	-0.303	-0.024	0.167
Organizational experience	-0.016	-0.209	1.000	-0.025	-0.058	-0.383	0.867	0.212	0.155
Organizational focus	-0.011	-0.038	-0.025	1.000	-0.016	-0.057	0.008	-0.182	-0.159
Prior-art citations	0.502	0.817	-0.058	-0.016	1.000	0.140	-0.107	-0.040	0.114
Recombination degree	0.014	0.470	-0.383	-0.057	0.140	1.000	-0.492	0.070	0.120
Field experience	-0.027	-0.303	0.867	0.008	-0.107	-0.492	1.000	0.112	0.094
Same-year patents	-0.018	-0.024	0.212	-0.182	-0.040	0.070	0.112	1.000	0.300
Firm age	0.005	0.167	0.155	-0.159	0.114	0.120	0.094	0.300	1.000

Correlation matrix for the main numeric variables.

G Appendix G: Yearly Audit

Period	V1	V2	V3	V4	V5
1986.0000	7306.0000	-1.9726	-0.7782	4.3848	0.0549
1987.0000	7920.0000	-1.8485	-0.5598	4.4559	0.0561
1988.0000	8927.0000	-1.7362	-0.4405	4.4998	0.0543
1989.0000	9876.0000	-1.7341	-0.3318	4.7117	0.0504
1990.0000	10409.0000	-1.7150	-0.2404	5.1830	0.0524
1991.0000	10950.0000	-1.6663	-0.1752	5.6408	0.0522
1992.0000	11688.0000	-1.5099	-0.1298	6.1589	0.0543
1993.0000	12054.0000	-1.3694	-0.1058	6.4200	0.0500
1994.0000	13712.0000	-1.2335	-0.0605	7.0965	0.0508
1995.0000	16332.0000	-0.8868	0.0240	7.2316	0.0500
1996.0000	15923.0000	-0.9553	0.0207	7.6327	0.0497
1997.0000	18799.0000	-0.8085	0.0155	7.7720	0.0536
1998.0000	18164.0000	-0.6502	-0.0664	7.6026	0.0506
1999.0000	19543.0000	-0.3911	-0.0794	7.2368	0.0522
2000.0000	22175.0000	-0.0682	-0.1024	6.7181	0.0512
2001.0000	22695.0000	0.0757	-0.1184	5.4978	0.0525
2002.0000	22461.0000	0.2464	-0.0957	5.3638	0.0485
2003.0000	22831.0000	0.4267	-0.0539	5.1639	0.0487
2004.0000	21790.0000	0.7716	-0.0321	5.3452	0.0492
2005.0000	21793.0000	0.7341	-0.0008	5.2593	0.0514
2006.0000	20534.0000	0.9402	0.0381	5.5693	0.0491
2007.0000	22148.0000	1.1377	0.0755	5.7011	0.0527
2008.0000	23454.0000	0.7233	0.1670	5.0370	0.0540
2009.0000	17971.0000	1.1788	0.1283	5.1727	0.0525
2010.0000	18599.0000	1.1069	0.1798	7.3907	0.0501
2011.0000	19134.0000	0.9084	0.1868	8.3084	0.0498

Period	V1	V2	V3	V4	V5
2012.0000	21603.0000	0.8664	0.3699	7.4903	0.0504
2013.0000	16415.0000	0.7798	0.4202	7.2127	0.0490
2014.0000	9657.0000	0.7923	0.5188	7.0003	0.0490
2015.0000	3791.0000	0.5775	0.8984	5.9330	0.0522
2016.0000	539.0000	-0.4169	1.3178	3.7625	0.0482

Year-by-year sample audit.

H Appendix H: Grouped Outcome Patterns

Quantile	N	MeanRegressor	MeanOutcome
1.0000	229109.0000	-2.5926	4.9416
2.0000	76899.0000	-1.0495	5.3861
3.0000	88598.0000	0.3530	5.8387
4.0000	94587.0000	6.7691	10.3793

Quantile summary for the main regressor.

I Appendix I: Full Coefficient Table

term	estimate	std_error	z_value	p_value	model	vcov
breadth_c	0.0030	0.0158	0.1921	0.8477	poisson_baseline	cluster_gvkey_ap
org_recomb_exp_log_c	0.0706	0.0354	1.9940	0.0461	poisson_baseline	cluster_gvkey_ap
org_focus_c	-0.3609	0.3761	-0.9595	0.3373	poisson_baseline	cluster_gvkey_ap
n_prior_art_cites	0.0018	0.0008	2.2149	0.0268	poisson_baseline	cluster_gvkey_ap
recomb_degree	-0.4546	0.1934	-2.3506	0.0187	poisson_baseline	cluster_gvkey_ap
field_recomb_exp_log	-0.1139	0.0457	-2.4923	0.0127	poisson_baseline	cluster_gvkey_ap
total_patents_same_year	-0.0001	0.0001	-1.4188	0.1560	poisson_baseline	cluster_gvkey_ap
total_patents_until_prior_log	-0.1661	0.0839	-1.9802	0.0477	poisson_baseline	cluster_gvkey_ap
breadth_c:org_recomb_exp_log_c	-0.0022	0.0008	-2.6692	0.0076	poisson_baseline	cluster_gvkey_ap
breadth_c:org_focus_c	-0.0050	0.1059	-0.0474	0.9622	poisson_baseline	cluster_gvkey_ap
org_recomb_exp_log_c:org_focus_c	0.3219	0.1779	1.8097	0.0703	poisson_baseline	cluster_gvkey_ap
breadth_c:org_recomb_exp_log_c:org_focus_c	-0.1705	0.0702	-2.4276	0.0152	poisson_baseline	cluster_gvkey_ap
breadth_c	0.0147	0.0147	1.0030	0.3159	logit_top5	cluster_gvkey_ap
org_recomb_exp_log_c	0.0988	0.0442	2.2329	0.0256	logit_top5	cluster_gvkey_ap
org_focus_c	-0.8737	1.3579	-0.6434	0.5200	logit_top5	cluster_gvkey_ap
n_prior_art_cites	0.0036	0.0019	1.9145	0.0556	logit_top5	cluster_gvkey_ap
recomb_degree	-0.3229	0.1351	-2.3907	0.0168	logit_top5	cluster_gvkey_ap
field_recomb_exp_log	-0.1319	0.0343	-3.8398	0.0001	logit_top5	cluster_gvkey_ap
total_patents_same_year	-0.0001	0.0000	-4.7631	0.0000	logit_top5	cluster_gvkey_ap
total_patents_until_prior_log	-0.3869	0.1008	-3.8305	0.0001	logit_top5	cluster_gvkey_ap
breadth_c:org_recomb_exp_log_c	-0.0118	0.0023	-5.0773	0.0000	logit_top5	cluster_gvkey_ap
breadth_c:org_focus_c	-0.1306	0.3120	-0.4187	0.6754	logit_top5	cluster_gvkey_ap
org_recomb_exp_log_c:org_focus_c	0.0361	0.8998	0.0401	0.9680	logit_top5	cluster_gvkey_ap
breadth_c:org_recomb_exp_log_c:org_focus_c	-0.2765	0.2994	-0.9236	0.3557	logit_top5	cluster_gvkey_ap
breadth_c	-0.0099	0.0231	-0.4280	0.6686	logit_top1	cluster_gvkey_ap
org_recomb_exp_log_c	0.1380	0.0934	1.4777	0.1395	logit_top1	cluster_gvkey_ap

term	estimate	std_error	z_value	p_value	model	vcov
org_focus_c	-0.9087	3.6923	-0.2461	0.8056	logit_top1	cluster_gvkey_ap
n_prior_art_cites	0.0048	0.0022	2.2062	0.0274	logit_top1	cluster_gvkey_ap
recomb_degree	-0.4480	0.2543	-1.7618	0.0781	logit_top1	cluster_gvkey_ap
field_recomb_exp_log	-0.2116	0.0677	-3.1246	0.0018	logit_top1	cluster_gvkey_ap
total_patents_same_year	-0.0002	0.0001	-2.6951	0.0070	logit_top1	cluster_gvkey_ap
total_patents_until_prev_log	-0.5362	0.2074	-2.5852	0.0097	logit_top1	cluster_gvkey_ap
breadth_c:org_recomb_exp_log_c	-0.0132	0.0056	-2.3569	0.0184	logit_top1	cluster_gvkey_ap
breadth_c:org_focus_c	-0.1517	1.2533	-0.1211	0.9036	logit_top1	cluster_gvkey_ap
org_recomb_exp_log_c:org_focus_c	-0.5147	3.3074	-0.1556	0.8763	logit_top1	cluster_gvkey_ap
breadth_c:org_recomb_exp_log_c:org_focus_c	-0.6392	1.0331	-0.6187	0.5361	logit_top1	cluster_gvkey_ap

Full coefficient dump exported from the local estimation script.

J Appendix J: Missingness Audit

Variable	Missing	MissingShare
Inventive impact	0.0000	0.0000
Breadth	0.0000	0.0000
Organizational experience	0.0000	0.0000
Organizational focus	0.0000	0.0000
Prior-art citations	0.0000	0.0000
Recombination degree	0.0000	0.0000
Field experience	0.0000	0.0000
Same-year patents	0.0000	0.0000
Firm age	0.0000	0.0000

Missingness audit for the variables summarized in the manuscript.

K Appendix K: Coverage of the Most Observed Entities

Entity	Label	Observations	FirstPeriod	LastPeriod
6066.0000	INTERNATIONAL BUSINESS MACHS COR	83091.0000	1986.0000	2016.0000
5047.0000	GENERAL ELECTRIC CO	28950.0000	1986.0000	2016.0000
12141.0000	MICROSOFT CORP	26529.0000	1994.0000	2016.0000
5606.0000	HEWLETT PACKARD CO	24186.0000	1986.0000	2016.0000
7585.0000	MOTOROLA INC	15304.0000	1986.0000	2016.0000
1300.0000	ALLIED SIGNAL INC	14534.0000	1986.0000	2016.0000
10499.0000	TEXAS INSTRUMENTS INC	14212.0000	1986.0000	2016.0000
4194.0000	EASTMAN KODAK CO	12898.0000	1986.0000	2010.0000
7435.0000	MINNESOTA MINING & MFG CO	12696.0000	1986.0000	2016.0000
4087.0000	DU PONT E I DE NEMOURS & CO	11004.0000	1986.0000	2016.0000
4839.0000	FORD MOTOR CO DEL	11003.0000	1986.0000	2016.0000
1690.0000	APPLE COMPUTER INC	10635.0000	1986.0000	2016.0000
6266.0000	JOHNSON & JOHNSON	10285.0000	1986.0000	2016.0000
2285.0000	BOEING CO	10165.0000	1986.0000	2016.0000

Entity	Label	Observations	FirstPeriod	LastPeriod
10983.0000	UNITED TECHNOLOGIES CORP	9824.0000	1986.0000	2016.0000
8762.0000	PROCTER & GAMBLE CO	8712.0000	1986.0000	2016.0000
4060.0000	DOW CHEMICAL CO	7294.0000	1986.0000	2016.0000
1704.0000	APPLIED MATERIALS INC	6975.0000	1995.0000	2016.0000
5073.0000	GENERAL MOTORS CORP	6710.0000	1986.0000	2008.0000
4503.0000	EXXON CORP	6538.0000	1986.0000	2016.0000

Top entities by local observation count.

L Appendix L: Source Inventory Note

Source Inventory - paper: Ghosh, Martin, Pennings, and Wezel (2014) - source title: Ambition Is Nothing Without Focus: Compensating for Negative Transfer of Experience in R&D - source file: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/Project 1_ Ghosh et al. 2014 replication/Ghosh-Martin-Pennings-Wezel-OS-25-(2)-2014.pdf‘ ## Paper-First Reading Protocol - Step 1: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/Project 1_ Ghosh et al. 2014 replication/Ghosh-Martin-Pennings-Wezel-OS-25-(2)-2014.pdf‘ — needed for: Read the original paper before touching the local estimates. — exists: n/a — status: workflow step - Step 2: ‘Ambition Is Nothing Without Focus: Compensating for Negative Transfer of Experience in R&D‘ — needed for: Extract the source paper’s theoretical logic, sample definition, variable definitions, and estimand. — exists: n/a — status: workflow step - Step 3: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/Data_Analysis/code/Regression/output/gvkey_clean_reproduction — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/ghosh-martin-pennings-wezel-2014-ambition-focus/results/ghosh_2014_model_coefficients.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/ghosh-martin-pennings-wezel-2014-ambition-focus/results/ghosh_2014_sample_summary.csv‘ — needed for: Match each required empirical object to a local executable file and verify that the file exists. — exists: n/a — status: workflow step - Step 4: ‘The manuscript, markdown notes, and PDF are generated from the verified file set.‘ — needed for: Write the replication paper only from the checked local files and calculated outputs. — exists: n/a — status: workflow step ## Local files - Source article: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/Project 1_ Ghosh et al. 2014 replication/Ghosh-Martin-Pennings-Wezel-OS-25-(2)-2014.pdf‘ — needed for: Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices — exists: Yes — status: Direct source - Analysis panel: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/Data_Analysis/code/Regression/output/gvkey_clean_reproduction_dataset.csv‘ — needed for: Estimate the local model and regenerate descriptive statistics — exists: Yes — status: Executable local dataset - Coefficient export: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/ghosh-martin-pennings-wezel-2014-ambition-focus/results/ghosh_2014_model_coefficients.csv‘ — needed for: Re-

cover the reported local coefficients and p-values — exists: Yes — status: Executable local result file - Sample summary: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/ghosh-martin-pennings-wezel-2014-ambition-focus/results/ghosh_2014_sample_summary.csv‘ — needed for: Audit sample construction and raw grouped moments — exists: Yes — status: Executable local result file

M Appendix M: Study Map Note

Study Map - research question: The source paper asks whether broad recombination becomes less productive when it leans on familiar routines, unless organizational focus offsets that negative transfer. - local design: The local rerun uses the clean GVKEY-anchored utility-patent panel already built in the workspace and keeps the patent-level interaction structure of the original paper. - estimand: A patent-level count model with interactions among breadth, organizational recombination experience, and organizational focus, plus firm and year controls. - headline estimate: '-0.0022' with 'p = 0.0076'. ## Decision Audit - direct source fact: The interaction structure is the theoretical core. - direct source fact: Forward citations measure impact. - direct source fact: The paper works at the patent level. local replication decision: The local rerun keeps the same interaction logic on the clean GVKEY-based panel. local replication decision: The local impact measure is the executable citation outcome already stored in the panel. local replication decision: The sample is narrowed to active S&P 500 utility patents with unambiguous assignment.

N Appendix N: Data Design Note

Data Design - Clean GVKEY-anchored S&P 500 utility-patent panel from the local replication pipeline. - PatentView-derived application year and citation outcomes. - CRSP-Compustat linked firm-year filters and sector information already present in the workspace. - panel file used: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/Data_Analysis/code/Regression/output - coefficient file used: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/ghosh-martin-pennings-wezel-2014-ambition-focus/results/ghosh_2014_m - sample summary used: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/ghosh-martin-pennings-wezel-2014-ambition-focus/results/ghosh_2014_sample_summary.csv‘ - observations: ‘489,193‘ - entities: ‘200‘ ## Required Data And Local File Mapping - Source article: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/Project 1_ Ghosh et al. 2014 replication/Ghosh-Martin-Pennings-Wezel-OS-25-(2)-2014.pdf‘ — needed for: Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices — exists: Yes — status: Direct source - Analysis panel: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/Data_Analysis/code/Regression/output/gvkey_clean_reproductio — needed for: Estimate the local model and regenerate descriptive statistics — exists: Yes — status: Executable local dataset - Coefficient export: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/ghosh-martin-pennings-wezel-2014-ambition-focus/results/ghosh_2014_model_coefficients.csv‘ — needed for: Recover the reported local coefficients and p-values — exists: Yes — status: Executable local result file - Sample summary: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/ghosh-martin-pennings-wezel-2014-ambition-focus/results/ghosh_2014_sample_summary.csv‘ — needed for: Audit sample construction and raw grouped moments — exists: Yes — status: Executable local result file

O Appendix O: Derivations Note

Derivations And Estimands - The local rerun strongly preserves the negative breadth-by-experience interaction and the negative three-way interaction with organizational focus. - A patent-level count model with interactions among breadth, organizational recombination experience, and organizational focus, plus firm and year controls. - model equations are reproduced in ‘paper/main.tex‘. ## Variable Map - Inventive impact -¿ ‘inventive_impact‘ : Use the local cleaned citation outcome already stored in the regression dataset. - Breadth -¿ ‘breadth_c‘ : Carry forward the centered breadth variable from the local regression output. - Organizational experience -¿ ‘org_recomb_exp_log_c‘ : Carry forward the centered log experience variable from the same panel. - Organizational focus -¿ ‘org_focus_c‘ : Use the contemporaneous focus measure already reconstructed in the local pipeline. - Controls -¿ ‘n_prior_art_cites; recomb_degree; field_recomb_exp_log; total_patents_same_year; firm_age‘ : Reuse the local patent-level controls and fixed effects.

P Appendix P: Replication Decisions Note

Replication Decisions - The local clean sample is paper-comparable rather than guaranteed identical to the archival source frame. - Patent-firm matching and sample filters follow the executable local pipeline. - Interpretation should emphasize mechanism recovery more than exact coefficient duplication. - direct source facts are separated from local replication decisions in the manuscript. - no coefficient or table is fabricated. ## Source Facts Versus Local Decisions - source: The interaction structure is the theoretical core. - source: Forward citations measure impact. - source: The paper works at the patent level. decision: The local rerun keeps the same interaction logic on the clean GVKEY-based panel. decision: The local impact measure is the executable citation outcome already stored in the panel. decision: The sample is narrowed to active S&P 500 utility patents with unambiguous assignment.

Q Appendix Q: Local Execution Status Note

Local Execution Status - batch date: ‘2026-04-04‘ - status: long-form journal-style manuscript generated from local result files - current paper: ‘ghosh-martin-pennings-wezel-2014-ambition-focus‘

R Appendix R: Cleaned Output Tables Note

Cleaned Output Tables - paper-first reading protocol - required-data and file-manifest table - sample reconstruction table - local source inventory table - source facts versus replication decisions table - variable mapping table - key coefficient table - full coefficient table - descriptive statistics - correlation matrix - yearly audit - missingness audit - top-entity coverage table - quantile summary

S Appendix S: Build Notes

Build Notes

- batch date: ‘2026-04-04‘ - manuscript style: long-form journal-style replication draft - evidence rule: only processed local datasets and actual local estimates are used - workflow rule: read the source paper first, then map required datasets to verified local files, then write the paper from those files - skill path guiding the structure: ‘C:/Users/bai/.codex/skills/write-replication-journal-github/SKILL.md‘

T Appendix T: Manuscript Metadata

Field	Value
Paper	Ghosh, Martin, Pennings, and Wezel (2014)
Source title	Ambition Is Nothing Without Focus: Compensating for Negative Transfer of Experience in R&D
Authors	Anindya Ghosh, Xavier Martin, Johannes M. Pennings, and Filippo Carlo Wezel
Outlet	Organization Science
Batch date	2026-04-04
Rows	489,193
Entities	200
Headline term	breadth x org_recomb_exp_log
Headline estimate	-0.0022
Headline p-value	0.0076
Panel path	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/Data_Analysis/code/Regression/output/gvkey_clean_reproduction_dataset.cs

Metadata block for GitHub publishing and future reruns.

U Appendix U: Replication Summary Note

Research question: The source paper asks whether broad recombination becomes less productive when it leans on familiar routines, unless organizational focus offsets that negative transfer. Local design: The local rerun uses the clean GVKEY-anchored utility-patent panel already built in the workspace and keeps the patent-level interaction structure of the original paper. Estimand: A patent-level count model with interactions among breadth, organizational recombination experience, and organizational focus, plus firm and year controls. Local result focus: The local rerun strongly preserves the negative breadth-by-experience interaction and the negative three-way interaction with organizational focus. Comparison note: This is the strongest executable match in the current batch. The negative breadth-by-experience interaction survives clearly, and the three-way interaction with focus remains negative and statistically meaningful, which means the central mechanism of the source paper is recovered in the local panel.

References

[1] Anindya Ghosh, Xavier Martin, Johannes M. Pennings, and Filippo Carlo Wezel. Ambition is nothing without focus: Compensating for negative transfer of experience in r&d. *Organization Science*, 25(2):572–590, 2014.